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SECOND SEMESTER M.Tech UNIVERSITY EXAMINATION, JUNE – 2019

AIRPLANE STABILITY AND CONTROL

Time:3 Hours **Maximum Marks:100**

Instructions:

1. Answer **FIVE** full Questions.
2. Questions numbers one and two are compulsory
3. Missing data may be suitably assumed.

1. a. Explain the following- **10**

- **i.** Meaning of stability and types of stability
- **ii.** Stick free neutral point
- **iii.** Condition for longitudinal static stability
- **iv.** Phugoid motion
- **v.** spin

b. An aircraft is undergoing steady and level flight with zero bank and side slip. Due to a sudden small disturbance, explain the different types of motion possible (with neat diagram) following the disturbance. **10**

2. a. Define and Explain the significance of Rouths criteria **08**

b. Drive an expression for the tail contribution to the pitching moment of an aircraft, if i_w is the wing setting angle and it is the tail setting angle. **12**

3. a. The moment at the C. G. due to wing fuselage is given by $C_{m_{cg,fw}} = -0.05 - 0.0035 \propto$. **15**

The following wing and tail data are given-

Wing: $S_w = 178\ ft^2$, $X_{cg}/c = 0.1$, $b_w = 35.9$ ft, $AR_w = 7.3$, $c_w = 5.0$ ft, $C_{L\alpha wf} = 0.07$/deg., $i_w = 2.0$ deg., $C_{L\alpha=0} = 0.26$

Tail: $l_t = 14.75$ ft, $\eta = 1$, $AR_t = 4.85$, $C_{L\alpha t} = 0.073$/deg.

Estimate the horizontal tail area and tail incidence angle i_t, so that the aircraft has following pitching moment characteristics- $C_{m_{cg}} = 0.15 - 0.025 \propto$

b. Explain yaw and roll stability and write down the necessary condition for yaw and roll stability **05**

OR

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4 a. Derive the expression for elevator angle to trim. 08

b. Derive the lateral equations of motion for an aircraft based on small perturbation theory. The lateral equations or motion in body fitted axes system is given by 12

$$F_y = m[\dot{v} + ru - pw]$$

$$L = [\dot{p} I_{xx} - \dot{r} I_{xz} + qr(I_{zz} - I_{yy}) - pqI_{xz}],$$

$$N = [-\dot{p} I_{xz} + \dot{r} I_{zz} + pq(I_{yy} - I_{xx}) - qrI_{xz}]$$

5. a. Describe different types of axes system used for study of airplane stability and control. 10

b. Explain in details how the forces acting on an airplane is modeled for longitudinal stability analysis. What will happen if the plane of symmetry is considered? 10

OR

6. a. What are unsymmetric disturbance and write the aerodynamic derivatives for symmetric disturbances and derive the expression for m_w? What will happen to aerodynamic derivatives if plane of symmetry is considered? 10

b. Explain in detail phugoid motion and short period oscillations. Write down the corresponding approximations for each types of motion. 10

7. a. Write short notes on 09

i. Spin

ii. Dutch roll

iii. Spiral divergence

b. Obtain the solution of lateral quartic $A1\lambda^4 + B1\lambda^3 + C1\lambda^2 + D1\lambda + E1 = 0$, where A1= 0.968, B1=5.72, C1=20.3, D1=97.2 and E1=21.3. 11

OR

8. a. Explain the possible roots for longitudinal quartic and corresponding motion of an airplane after a symmetric disturbance. 10

b. Explain the consequence of plane of symmetry in an airplane. 10